

Aahan Ghode

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SUMMARY

Driven and resourceful Computer Engineering student with experience in finite element analysis, data-driven intelligence, and scientific computing, skilled in C++, Python, and MATLAB, and eager to pursue opportunities in embedded systems and hardware-software integration.

WORK EXPERIENCE

Simulation Engineer Jul 2026 — Present
Sunnybrook Research Institute Toronto, ON

- Modeled OnScale FEA simulations for radially-poled PZT-4 ring transducers, optimizing acoustic beam focus across 5+ catheter geometry iterations
- Engineered MATLAB pipelines to parse binary time-histories, accelerating post-processing speeds
- Mapped spatial pressure distributions to maximize acoustic energy delivery while preventing cavitation damage
- Calculated piezoelectric mode purity from displacement fields, confirming clean length-mode excitation
- Analyzed dynamic mechanical stress distributions to identify standing wave peak stress, establishing safe drive voltage limits

Lead Assistant Teacher Jun 2023 — Jun 2026
Spirit of Math Schools Inc. Oakville, ON

- Taught advanced mathematics curricula to 40+ Grade 9–12 students
- Mentored 15+ students weekly on advanced coursework, improving individual topic mastery
- Participated in weekly administrative reviews with senior faculty, providing actionable feedback that refined 100% of student progress evaluations
- Co-developed personalized academic learning plans with lead instructors, increasing student retention

Full Stack Developer Intern Nov 2024 — May 2025
Vertex Dimension Remote

- Developed custom front-end components and integrated 3rd-party Shopify apps using HTML/CSS, improving page load speeds across 3 client sites
- Supported 2 project managers across 10+ client technical meetings, translating business requirements into precise engineering specifications
- Optimized Search Engine Optimization (SEO) structures and site navigation flow, increasing client e-commerce conversion rates

Database Administrator Intern Aug 2023 — Oct 2023
RS Tec Systems Inc. Oakville, ON

- Engineered an automated resume parser capable of batch processing 200+ applicant documents simultaneously
- Parsed and filtered unstructured resume data across 3 file formats (.pdf, .docx, .doc), reducing manual screening time
- Collaborated with 2 senior administrators to review feedback, implementing security enhancements
- Integrated Google Sheets API to automate spreadsheet syncs, eliminating manual data entry steps
- Designed a responsive Flask and Bootstrap UI, streamlining database management and administrative record entry for internal users

SKILLS

Programming Languages: Python, C++, Java, Node.js, Data structures and algorithms
Back-end: Flask, Django, REST APIs
Front-end: React, JavaScript, HTML, CSS, Next.js
Other: Linux, GitHub, MATLAB, Microsoft Suite (Word, Outlook, Excel, PowerPoint)
Languages: English, Marathi, French

HONOURS AND AWARDS

Jun. 2026	Minister’s Certificate of Recognition for Community Involvement (200+ Hours)	<i>Ontario Ministry of Education</i>
May. 2026	Schulich Leader Scholarship (\$120K)	<i>University of Toronto</i>
Nov. 2025	Certificate of Distinction (Top 25%) CSMC	<i>University of Waterloo</i>
Apr. 2025	Certificate of Distinction (Top 25%), Hypatia Math Contest	<i>University of Waterloo</i>
Jul. 2024	AP Scholar (Scored 3 or higher on 3 or more AP Exams)	<i>College Board</i>

EDUCATION

University of Toronto <i>Bachelor of Applied Science, Computer Engineering</i>	<i>Toronto, ON</i> <i>Sept 2026 — Apr 2031</i>
Iroquois Ridge High School <i>Ontario Secondary School Diploma, Clustered Program</i>	<i>Oakville, ON</i> <i>Sept 2022 — Jun 2026</i>